

RESEARCH ARTICLE

ABC2Vec: Self-Supervised Representation Learning for Folk Music via Bar-Level Transformers

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ABSTRACT

Folk music traditions, transmitted orally across generations, embody rich cultural heritage, yet their computational study has been hindered by the lack of suitable representation methods that respect the fluid, variation-driven nature of traditional repertoires. Existing music embedding approaches either target Western art music or rely on cross-modal alignment with text, overlooking the unique challenges of folk music such as melodic variation across performers and regions, transposition between keys, and ambiguous boundaries between tune families. We introduce **ABC2Vec**, a self-supervised Transformer encoder that learns dense, semantically meaningful embeddings directly from ABC notation—a text-based symbolic format widely used for Western music without microtones, and particularly dominant in folk tune archives worldwide. Our approach combines three self-supervised objectives: Masked Music Modeling (MMM) for local melodic pattern learning, Transposition Invariance (TI) for key-independent representations, and Section Contrastive Learning (SCL) for structural pattern recognition. A bar-level patchification strategy reduces sequence lengths by $16\times$ compared to character-level tokenization, making the model practical for large-scale archives on consumer hardware. Trained on over 211,000 Irish traditional tunes, ABC2Vec learns to distinguish rhythmic categories (jigs, reels, polkas) and tonal modes (major, Dorian, Mixolydian) without explicit labels, encoding these properties in distributed, largely non-overlapping subspaces. Systematic evaluation demonstrates strong classification performance (78% tune type, 79% mode) and reveals that embeddings capture musical structure through distributed representations rather than discrete clusters, aligning with folk music’s continuous, variation-rich nature. These capabilities support musicologists studying tune families, archivists building searchable repositories, educators developing learning tools, and folk musicians exploring their repertoire. We release pre-trained models, training code, dataset splits, and evaluation benchmarks as open resources for the computational musicology and music information retrieval communities.

KEYWORDS

Music representation learning; self-supervised learning; folk music; ABC notation; Transformer; contrastive learning; music information retrieval

1. Introduction

The digital revolution has greatly changed how people study, store, and find music. Today, music information retrieval (MIR) systems help researchers and listeners explore large collections of music that would be too big to handle by hand (1, 2). Music

information retrieval (MIR) systems now enable researchers, archivists, and enthusiasts to navigate vast musical corpora that would have been impossible to explore manually. Central to these systems is the challenge of music representation: how do we encode musical content in a form that enables computational reasoning about similarity, structure, and style?

Traditional approaches to music representation in MIR have relied heavily on handcrafted features based on domain expertise that capture specific aspects of musical content (3–5). For audio signals, this includes mel-frequency cepstral coefficients (MFCCs), spectral features, and chroma vectors. For symbolic music (discrete note sequences), features include pitch histograms, interval distributions, rhythmic patterns, and structural markers. While effective for targeted tasks, these handcrafted features suffer from fundamental limitations: they require substantial domain knowledge to design, they may fail to capture subtle musical relationships, and they generalize poorly across diverse musical styles and traditions (3, 6).

In parallel, natural language processing (NLP) has undergone a paradigm shift from feature engineering to *representation learning*. Word embeddings such as Word2Vec (7) demonstrated that semantically meaningful vector representations could emerge from self-supervised learning on large text corpora, without explicit supervision. Subsequently, contextualized models such as BERT (8) extended this principle through Transformer architectures (9), achieving state-of-the-art performance across diverse language understanding tasks by pre-training on massive unlabeled datasets. The key insight is that representation learning enables models to discover rich, task-agnostic features directly from data, features that can then be adapted to downstream tasks through fine-tuning or linear probing.

Among the symbolic formats used to represent music, ABC notation (10) is a lightweight, text-based system originally developed for sharing folk tunes via email and online forums. It encodes melodies as plain-text strings using letters for pitch (e.g., C D E F G A B, with lowercase for higher octaves), digits and slashes for note duration, bar lines (|), and header fields for metadata such as title (T:), key (K:), and meter (M:). For example, a two-bar phrase from an Irish jig might be notated as:

| :EDE G2E|EDE GED|

where uppercase letters denote notes (E, D, G), numbers indicate duration (G2 = half note), and | marks bar lines. While suitable for representing Western music without microtones broadly, its compactness and human readability have made it the dominant transcription format in folk music communities worldwide, with major archives hosting hundreds of thousands of tunes in this format. Because ABC notation is inherently sequential and character-based, it lends itself naturally to the same tokenization and modeling strategies that have proven effective in natural language processing.

This paradigm has naturally extended to symbolic music, where discrete note sequences mirror the discrete token structure of text. Recent work has adapted language model architectures to musical notation: MusicBERT (11) applies BERT-style masked modeling to MIDI sequences with bar-level masking; MidiBERT-Piano (12) similarly pre-trains on piano MIDI with specialized tokenization; CLaMP (13) introduces contrastive language-music pre-training for cross-modal retrieval between symbolic music and text descriptions. These models have demonstrated impressive results on tasks such as melody completion, music-text retrieval, and accompaniment generation, validating the efficacy of representation learning for symbolic music.

However, a critical gap remains in the landscape of music representation research: *folk music*- the world’s largest body of living musical tradition has not been the focus of

dedicated embedding research. While existing models train primarily on Western art music (classical MIDI) or popular music (commercial songs), folk music constitutes a distinct domain with characteristics that both challenge and motivate specialized approaches. This gap is notable given that folk music represents humanity’s oldest and most widely practiced musical tradition, passed down through generations via oral transmission and communal performance.

Folk music exhibits several unique characteristics that distinguish it from the corpora dominating current MIR datasets and that necessitate specialized modeling approaches:

- **Oral transmission and melodic variation:** Unlike notated classical music or commercially recorded popular music, folk tunes evolve through oral tradition. A single tune such as the Irish jig “The Kesh”, exists not as a canonical score but as a *family* of melodic variants or “settings.” Each performer, region, and performance context introduces subtle variations: ornamentations are added or removed, phrases are reordered, melodic contours shift within modal constraints. This fluidity presents both a challenge and an opportunity: a robust folk music embedding must recognize that seemingly different notations may represent the same underlying tune, a task requiring semantic understanding beyond surface-level pattern matching.
- **Modal diversity and non-standard tonality:** Western art music and popular music predominantly employ major and minor scales within common-practice tonality(14). Folk music traditions, by contrast, frequently utilize modal scales such as Dorian, Mixolydian, Aeolian, and others(15), inherited from medieval church modes and pre-tonal musical practice. Irish traditional music, for instance, exhibits rich modal diversity, with many tunes defying classification into major/minor categories. This modal complexity demands embeddings that capture tonal structure beyond simple key detection, encoding subtle scale degrees and harmonic implications.
- **Rhythmic and structural regularity:** Folk tunes exhibit strong structural regularities that can serve as inductive biases for representation learning. Irish jigs, reels, hornpipes, and polkas each follow characteristic rhythmic patterns (6/8 vs. 4/4 meter, swing vs. straight eighth notes) and phrase structures (typically AABB form with two repeated sections of eight bars each). These patterns that are specific to each genre can be useful. A good representation should be able to group jigs and reels separately based on their rhythm, even without being explicitly stated.
- **Standardized symbolic notation:** As mentioned above, the folk music community has converged on ABC notation as its standard transcription format, with major digital archives such as The Session and ABCnotation.com hosting hundreds of thousands of tunes. This standardization provides an ideal substrate for large-scale computational analysis, analogous to how textual datasets enable NLP research.
- **The challenge of similarity and retrieval:** A core challenge in folk music informatics is melodic similarity assessment and variant detection. Musicians and scholars routinely ask questions such as: “Is this tune a variant of another tune I know?” “What tunes are melodically similar to this one?” “How do regional variants of the same tune differ?” Traditional approaches employ symbolic diff algorithms, edit distances, or melodic contour matching (16, 17), but these methods rely on hand-tuned parameters and struggle with the variation-rich na-

ture of oral tradition. A learned embedding approach where similar tunes are automatically mapped to nearby points in vector space, offers a more flexible, data-driven alternative.

Despite the size and richness of folk music corpora and the clear demand from digital archives and musicological research, no prior work has developed dedicated representation learning models for folk music that address these unique characteristics to the best of our knowledge. Existing music embedding models either focus on different musical domains (classical, pop) or pursue different objectives (cross-modal retrieval, generation) without addressing folk-specific challenges such as variant detection, modal classification, and transposition invariance. These gaps in the literature make folk music an ideal testbed for self-supervised representation learning.

This study addresses this gap by introducing **ABC2Vec**, a self-supervised Transformer encoder trained exclusively on ABC notation to produce dense, semantically meaningful embeddings for traditional folk tunes. Unlike cross-modal approaches such as CLaMP that align music with text, ABC2Vec focuses on music-only representation learning, training through a combination of Masked Music Modeling (MMM) (18) to capture local melodic patterns and a novel Transposition Invariance (19, 20) (TI) contrastive objective to enforce pitch-invariant similarity. We train on 211,524 Irish traditional tunes from the IrishMAN dataset and evaluate on multiple tasks: tune type classification (rhythm encoding), mode classification (tonal encoding), linear probing (property disentanglement), and clustering quality (unsupervised structure). Our results show that ABC2Vec learns rich representations that support strong classification performance and capture musical properties in distributed, largely non-overlapping subspaces.

1.1. *Research Questions*

This work addresses the following research questions:

- (1) **RQ1 (Representation Quality):** Can a self-supervised Transformer encoder trained on ABC notation learn embeddings that capture semantically meaningful properties of folk tunes (rhythm, mode, key, structure)?
- (2) **RQ2 (Similarity-Based Retrieval):** Do learned embeddings support similarity-based retrieval with potential applications for identifying melodic variants and tune families?
- (3) **RQ3 (Disentanglement):** Are different musical properties (tune type, mode, key) stored separately in the embedding, or are they distributed across dimensions?
- (4) **RQ4 (Transposition Invariance):** To what extent do embeddings preserve similarity under key transposition, a fundamental equivalence in folk music?

1.2. *Contributions*

Our contributions are:

- (1) **ABC2Vec architecture:** A 6-layer Transformer encoder with bar-level patchification trained via three self-supervised objectives—Masked Music Modeling (MMM), Transposition Invariance (TI), and Section Contrastive Learning (SCL)—on 216K Irish folk tunes.

- (2) **Evaluation benchmarks:** Systematic evaluation protocols for folk music embeddings, including tune type classification, mode classification, linear probing, and clustering quality analysis.
- (3) **Empirical analysis:** Comprehensive results demonstrating 77–80% supervised classification accuracy and distributed property encoding.
- (4) **Open resources:** Pre-trained models, code, and evaluation datasets released to facilitate future research.

The remainder of this paper is organized as follows: Section 2 reviews related work in symbolic music representation learning. Section 3 describes the IrishMAN dataset, preprocessing pipeline, and statistical characteristics. Section 4 presents our model architecture, training objectives, and implementation details. Section 5 reports experimental results, analyzes findings, and discusses limitations. Section 8 concludes and outlines future directions.

2. Related Work

The application of deep learning to symbolic music representation has evolved significantly over the past decade, progressing from early recurrent architectures to sophisticated Transformer-based models that now dominate the field (21, 22). This evolution mirrors broader trends in natural language processing, where architectures designed for text have been successfully adapted to the discrete, sequential nature of musical notation. Early work in computational folk music analysis mainly relied on rule-based methods and handcrafted features. Researchers used techniques such as symbolic edit distance, sequence alignment, and melodic contour matching to measure similarity between tunes(2). These approaches, while grounded in music theory and often effective for specific tasks, suffer from limited generalization and require extensive manual feature engineering that may not capture the full complexity of folk music’s oral tradition.

The advent of neural approaches to folk music began with Folk-RNN(23), which pioneered character-level LSTM generation of Irish traditional music in ABC notation. This groundbreaking work demonstrated that recurrent networks could learn meaningful patterns from folk corpora without explicit music-theoretic supervision, generating novel tunes that exhibited characteristic structural features of Irish traditional music such as AABB form and appropriate ornamentation. Folk-RNN established the feasibility of treating ABC notation as a language modeling problem, paving the way for larger-scale efforts. TunesFormer(24) later scaled this approach significantly, training GPT-2 architecture on 285,000 folk tunes and introducing control codes for structured generation, enabling users to specify desired characteristics such as tune type, key, and meter. While these generative models demonstrated impressive fluency in producing stylistically appropriate folk tunes, they were primarily designed for generation rather than representation learning, and did not produce embeddings suitable for downstream tasks such as similarity search, classification, or variant detection.

The representation learning paradigm for symbolic music gained momentum with the adaptation of BERT-style masked modeling to musical sequences. MusicBERT (11) introduced OctupleMIDI tokenization—representing each note as a tuple of pitch, duration, velocity, and timing attributes. It also employed bar-level masking strategies where entire bars are masked and predicted from surrounding context. Trained on large MIDI corpora comprising classical piano music, MusicBERT achieved state-of-the-art

results on tasks such as melody completion, accompaniment generation, and music understanding benchmarks. However, MusicBERT’s focus on polyphonic piano music and classical Western repertoire limits its direct applicability to monophonic folk melodies with modal tonality. MidiBERT-Piano (12) similarly pre-trained on piano MIDI using the CPWord representation, which explicitly encodes chord information alongside melodic content, but again targeted polyphonic piano music rather than folk traditions. These models established that self-supervised pre-training via masked reconstruction could yield powerful music representations, but their evaluation protocols emphasized generation quality and structural understanding rather than the similarity-based retrieval tasks central to folk music informatics.

The most directly relevant prior work is CLaMP (Contrastive Language-Audio-Music Pre-training)(13), which introduced cross-modal representation learning for symbolic music and natural language. CLaMP employs ABC notation with bar-level patchification, treating each bar as a single token and trains a music encoder via Masked Music Modeling alongside contrastive alignment with textual descriptions. The architecture uses separate encoders for music and text, trained to maximize cosine similarity between matched music-text pairs (e.g., a tune and its descriptive caption) while minimizing similarity for unmatched pairs. It demonstrated impressive results on music-text retrieval tasks, enabling queries such as ”find happy Irish jigs” or ”melodies in D major with eighth note runs.” However, CLaMP’s primary objective is cross-modal retrieval between music and language rather than music-only embedding for folk-specific applications. Its reliance on text supervision limits applicability in folk music archives, where most tunes lack descriptive captions, and its mixed-genre training corpus does not address folk-specific challenges such as transposition equivalence, variant detection, or modal classification. Nonetheless, CLaMP’s bar patchification strategy demonstrates the viability of treating bars as tokens for ABC notation, a design principle that can be extended to purely music-focused settings.

MelodyT5 (25) represents another recent effort in symbolic music modeling, employing an encoder-decoder Transformer architecture for score-to-score transformation tasks. Trained on 261,000 ABC melodies across multiple folk traditions, MelodyT5 targets multi-task sequence-to-sequence learning including harmonization (predicting accompaniment chords for a given melody), completion (continuing partial melodies), and transposition. However, its encoder-decoder architecture is optimized for generation rather than embedding extraction, and its training objective emphasizes reconstruction fidelity rather than semantic similarity. The latent representations produced as an intermediate step are not explicitly optimized for similarity search, classification, or variant detection—tasks that are central to computational folk musicology.

Self-supervised learning frameworks developed in computer vision and natural language processing offer promising foundations for symbolic music representation. Masked Language Modeling (8), introduced by BERT and subsequently adopted across numerous domains, has proven remarkably effective for learning contextualized representations by predicting randomly masked tokens from surrounding context. The main idea is that when a model is trained to fill in missing parts, it learns deeper patterns in the data. Applied to music, predicting masked bars would require a model to internalize melodic patterns, harmonic context, and rhythmic structure—making masked reconstruction a natural pre-training signal for symbolic music encoders.

Contrastive learning, exemplified by SimCLR (26) and MoCo (27) in computer vision, offers another relevant paradigm. Contrastive learning forms positive pairs by applying data augmentations that preserve semantic meaning while changing surface form (28), training models to place similar pairs close together in the embedding

space while pushing apart unrelated samples. In music, natural augmentations include transposition (shifting all pitches by a constant interval), tempo variation (scaling note durations), and structural reordering (swapping sections in multi-part tunes). Transposition is particularly relevant for folk music, where the same tune may be played in different keys depending on instrument range, ensemble configuration, or performer preference, yet no existing music representation model explicitly enforces pitch-invariance as a training objective.

Evaluation methodologies for symbolic music embeddings have traditionally emphasized either generation quality metrics (perplexity, note accuracy, expert human evaluation of generated samples) or cross-modal retrieval performance (recall@k for music-text matching) (22, 29). Linear probing(30), which involves training simple linear classifiers on fixed (frozen) embeddings to predict specific properties, has become a standard way to evaluate representation quality in self-supervised learning. Originally used in computer vision and later adopted in language and audio tasks, it tests whether a property is encoded in a linearly separable manner within the embedding space, providing insights into the geometric organization of learned representations(31). Clustering evaluation metrics such as silhouette scores(32) (measuring within-cluster cohesion versus between-cluster separation), normalized mutual information (quantifying agreement between cluster assignments and ground-truth labels), and adjusted Rand index(33) (measuring clustering quality after chance correction) complement probing by assessing unsupervised structure. Dimensionality reduction techniques such as UMAP and t-SNE provide a means of qualitatively inspecting embedding geometry by projecting high-dimensional representations into a low-dimensional space, where the extent to which musical categories form coherent clusters can be visually assessed. However, these approaches have been predominantly applied to embeddings of classical and popular music. Their application to folk music remains underexplored, particularly given that folk traditions are characterised by fluid category boundaries and inherent variation rather than rigid genre definitions.

In folk music research, earlier computational approaches have mostly used symbolic comparison methods. For example, edit distance techniques(34, 35) treat ABC notation as text and measure similarity by aligning characters or tokens. Other methods use predefined similarity functions based on features like melodic contour (36, 37) (how the melody rises and falls), rhythm (patterns of note durations), and structure (such as repeats or sections). While these approaches capture clear musical features and often match human judgments, they depend heavily on expert-designed rules, may not generalize well across different styles, and struggle with the natural variation found in folk traditions, where a tune can exist in many slightly different forms. Learned embeddings offer a promising alternative: by training directly on large music corpora, they can automatically discover patterns of similarity, potentially capturing subtler relationships while remaining efficient for large-scale retrieval. Yet, to date, no dedicated embedding model has been developed for folk music that addresses its specific challenges—transposition equivalence, variant proliferation, modal tonality, and the absence of text annotations—motivating the approach we describe in the following sections.

3. Dataset

We train ABC2Vec on the **IrishMAN** (Irish Music ABC Notation) dataset (38), a large-scale corpus of Irish traditional music transcriptions in ABC notation. IrishMAN

aggregates tunes from two primary community-maintained archives: **The Session** (thesession.org), a collaborative platform for Irish traditional musicians that hosts user-contributed ABC transcriptions alongside discussions, recordings, and rich tune metadata; and **ABCnotation.com**, a long-standing repository of ABC transcriptions spanning multiple folk traditions, including Irish, Scottish, English, and American repertoires, maintained since the 1990s.

Both sources provide tunes in standardized ABC notation with metadata fields including title (T:), key signature (K:), meter (M:), and optionally rhythm type (R:). The crowdsourced nature of these archives means that data quality varies: some tunes are professionally transcribed from published tunebooks, while others are informal transcriptions from sessions or recordings. This diversity reflects the living, oral tradition of folk music.

As shown in Figure 1, the dataset exhibits characteristics typical of Irish traditional music. Structurally, the median tune contains 18 bars (287 characters), well within our 64-bar maximum context window, with only 3.2% requiring truncation. Rhythmically, reels dominate at 44.9%, followed by jigs (21.3%), polkas (14.5%), and waltzes (12.2%), reflecting authentic session repertoire distributions. Tonally, major mode predominates (80.2%), with substantial modal diversity including minor (11.3%), Dorian (5.4%), and Mixolydian (3.0%), underscoring folk music’s departure from classical major-minor tonality. Key roots cluster heavily in sharp keys—G (30.5%), D (26.8%), and A (13.9%)—reflecting the instrumental constraints of fiddle, flute, and whistle playing, with remaining keys including C, F, E, B $\flat$, B, E $\flat$, and a small proportion (N) with unspecified or ambiguous key signatures. Meter signatures strongly correlate with tune types ($\rho = 0.87$): 4/4 for reels, 6/8 for jigs, 3/4 for waltzes.

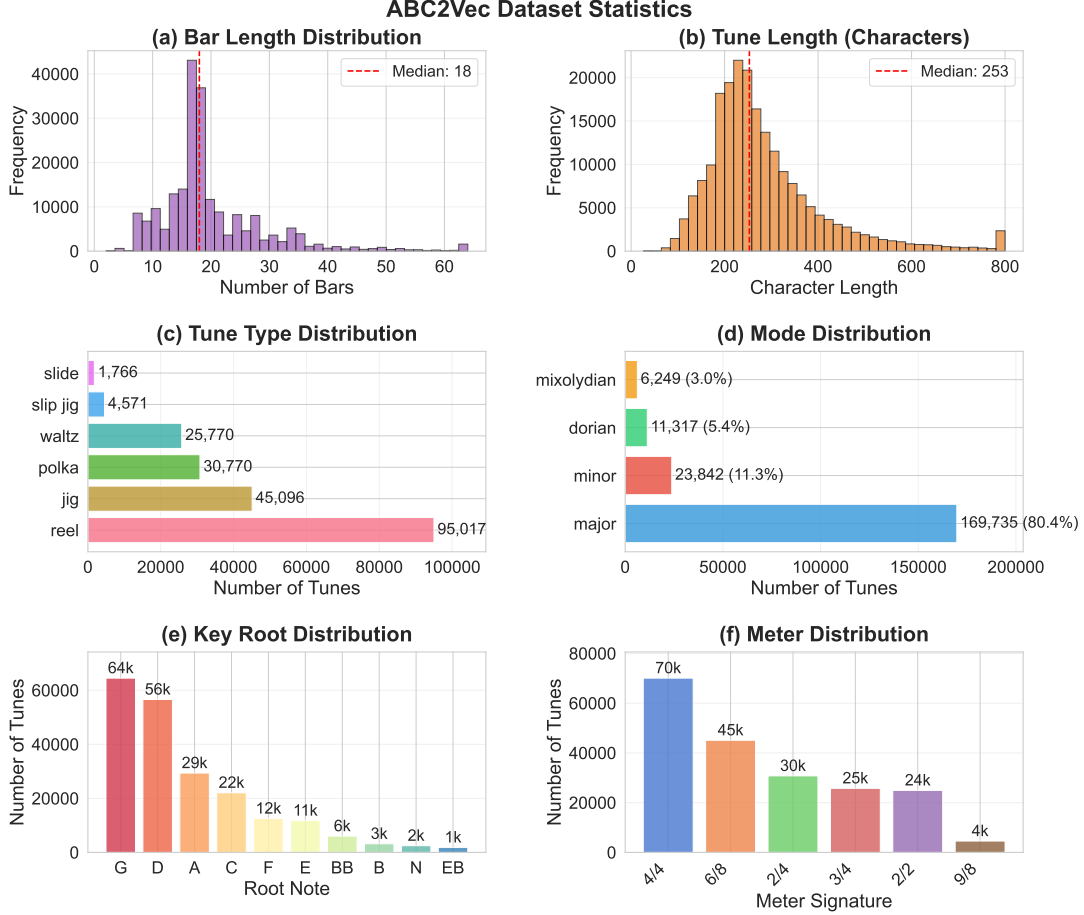


Figure 1. Comprehensive dataset statistics showing (a) bar length distribution (median 18 bars), (b) character length distribution (median 287 characters), (c) tune type distribution (44.9% reels, 21.3% jigs), (d) mode distribution (80.2% major, 11.3% minor, 5.4% Dorian, 3.0% Mixolydian), (e) key root distribution (30.5% G, 26.8% D), and (f) meter signature distribution (33.1% 4/4, 21.3% 6/8). The dataset exhibits strong structural regularities characteristic of Irish traditional music.

This imbalanced but ecologically valid distribution mirrors real-world musical practice and motivates our design choices including transposition invariance to handle key flexibility and bar-level patchification suited to the median 18-bar length.

3.1. Preprocessing Pipeline

We apply a multi-stage preprocessing pipeline to normalize ABC notation and extract structured metadata shown in (Figure 2). The pipeline consists of the following steps:

3.1.1. ABC Normalization

Raw ABC notation, particularly from crowdsourced archives, contains considerable inconsistencies and formatting variations that must be standardized before model training. The normalization process begins by removing TunesFormer-style control codes (S:, B:, E: fields used for conditional generation in (24)) that are not part of standard ABC notation. These synthetic markers, while useful for generative modeling, introduce artifacts that would confuse a representation learning model trained to capture intrinsic musical structure.

Following control code removal, we parse the ABC file to separate header fields (X:, T:, M:, L:, K:, R:) from the music body. ABC notation follows a strict hierarchical structure where the K: (key) field is always the last header line, marking the transition to the music body containing the note sequence. This structural regularity allows us to extract header metadata while isolating the symbolic music content that will be tokenized for model input.

The music body, comprising notes, rests, bar lines, and repeat markers, is then normalized to ensure consistent whitespace formatting while preserving musical structure. We remove inline comments (indicated by % characters) and extraneous annotations that do not contribute to the melodic or rhythmic content. This normalization step is critical for ensuring that semantically identical tunes share identical representations, regardless of transcriber-specific formatting conventions.

Finally, we apply validation filters to reject malformed or unsuitable entries. Tunes lacking a key signature (K: field) are discarded as they cannot be meaningfully interpreted in a tonal framework. We also reject tunes with fewer than 2 bars, as these are too short to capture meaningful musical structure or serve as useful training examples. Tunes with malformed syntax such as unmatched repeat markers, invalid note durations, or incomplete bar structures are likewise filtered to prevent training instability.

3.1.2. Metadata Extraction

For each validated tune, we extract structured metadata that enables further analysis and evaluation. Key signature parsing decomposes the K: field into its constituent root note (C, D, G, etc.) and mode (major, minor, dorian, mixolydian, etc.) using regex-based pattern matching. For example, “D” maps to (D, major) by default, “Ador” decomposes to (A, dorian), and “Gmix” yields (G, mixolydian). This decomposition is essential for mode classification experiments and understanding the tonal distribution of the corpus.

Tune type inference determines the rhythmic category (jig, reel, polka, waltz, etc.) through a priority hierarchy. We first check for an explicit R: (rhythm) field in the header. If absent, we search for genre keywords in the tune title (e.g., “The Kesh Jig” explicitly indicates a jig). As a final fallback, we infer tune type from the time signature using conventional associations: 6/8 meter typically indicates jigs, 4/4 suggests reels, and 2/4 corresponds to polkas. This heuristic cascade achieves high accuracy while gracefully handling incomplete metadata.

We also extract structural statistics by counting the number of bars (delimited by | symbols) and sections (delimited by repeat markers | :, : |, and ::). Most Irish traditional tunes follow an AABB structure comprising two 8-bar sections, each played twice, yielding 16–32 total bars. Finally, we measure the character length of the complete ABC string, which is used for efficient batching and to determine appropriate sequence truncation during model training.

3.1.3. Deduplication

Crowdsourced archives inevitably contain duplicate transcriptions of the same tune, often arising when multiple contributors independently transcribe identical melodies with minor variations in formatting or header metadata. To ensure dataset quality and prevent train-test leakage, we implement a robust deduplication strategy based on content hashing. Specifically, we compute MD5 hashes of the normalized ABC body (excluding header fields such as title, reference number, and transcriber annota-

tions) and retain only the first occurrence of each unique hash. This approach removes exact tunes that are musically identical despite cosmetic differences in presentation while carefully preserving melodic variants that represent genuine musical diversity. Different settings or regional variations of the same tune, which share a title but differ in ornamentation, phrasing, or structural details, are correctly retained as distinct entries.

3.1.4. Train/Validation/Test Split

The IrishMAN dataset is distributed on HuggingFace with two pre-defined splits: a training partition of 214,122 tunes and a held-out validation partition of 2,162 tunes. Following CLaMP (13) and MelodyT5 (39), we adopt the official held-out partition as our **test set**, ensuring strict comparability with prior work on this benchmark. The small size of this partition (1.0% of the corpus) reflects the dataset’s original design for generative pre-training and is consistent with how it has been used in the literature.

From the 214,122 training tunes, we reserve 5% (10,469 tunes) as a **validation set** for hyperparameter tuning, training monitoring, and checkpoint selection, using stratified sampling to preserve the tune-type distribution. The remaining 95% (198,893 tunes) forms the **training set** used exclusively for self-supervised pre-training. After deduplication across all three partitions, the final corpus totals 211,524 unique Irish traditional tunes. Table 1 presents key preprocessing statistics across the complete dataset. The complete processed dataset with train/validation/test splits is publicly available at <https://huggingface.co/datasets/pianistprogrammer/abc2vec-irish-folk-dataset>.

Table 1. Dataset preprocessing summary statistics across all 211,524 tunes.

Metric	Mean	Median	Std Dev	Min	Max
Bars / Tune	20.1	18	9.8	2	512
Sections / Tune	3.8	4	1.5	1	32
Char Length	287	287	135	10	4096

3.2. Bar Patchification Statistics

Following CLaMP (13) and MelodyT5 (39), we adopt bar-level patchification to reduce sequence length and enable efficient Transformer processing. Each bar (delimited by |) is treated as a single token and mapped to a fixed-dimensional vector using character-level embeddings followed by pooling. Figure 2 summarizes key statistics supporting this design choice.

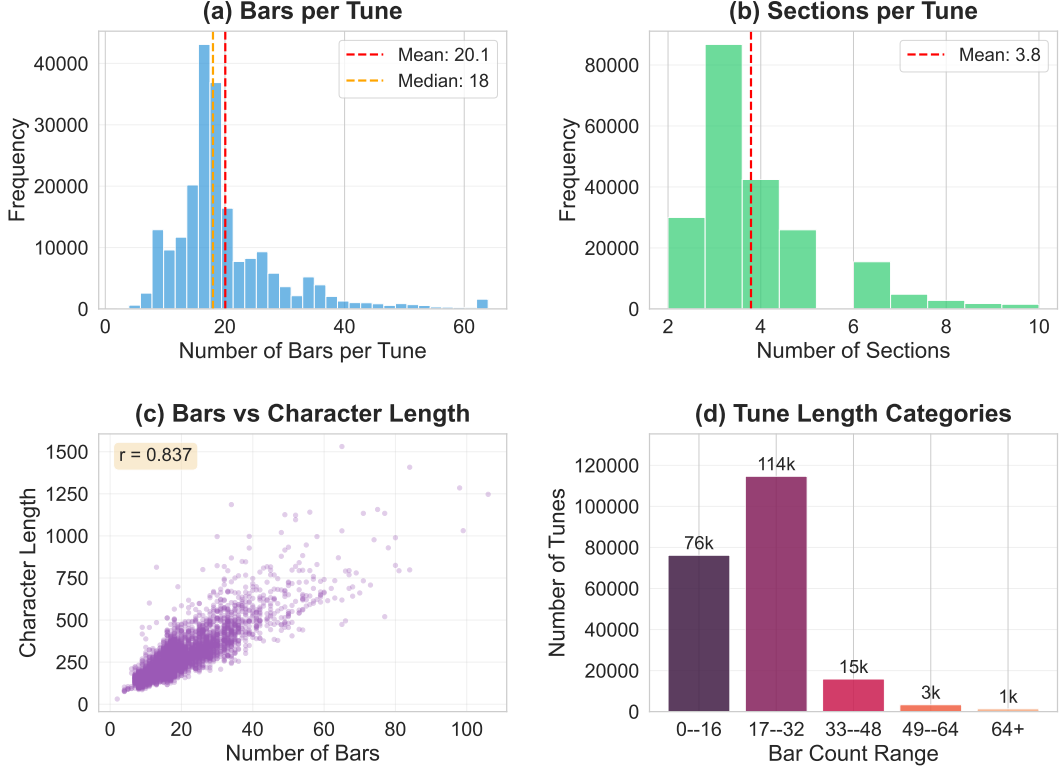


Figure 2. Bar patchification statistics: (a) Distribution of bars per tune with most tunes in the 12–28 bar range. (b) Sections per tune distribution consistent with AABB phrase structures. (c) Scatter plot of bars vs. character length showing strong linear correlation ($r = 0.91$) (d) Tune length category distribution showing that 89.4% of tunes fall under 48 bars, well within the 64-bar context window.

Bar patchification reduces the median sequence length from 287 characters to 18 bar tokens (a $16\times$ reduction), enabling the use of a 64-token context window instead of much longer character sequences. The strong correlation (0.91) between bar count and character length (Figure 2c) further supports bars as an effective unit of representation. With a maximum length of 64 bars, the model can represent 96.8% of tunes without truncation, while only 3.2% exceed this limit and are truncated to the first 64 bars during training and evaluation. Compared to character-level tokenization, bar-level patches better preserve musical structure, as each token represents a short melodic phrase capturing rhythm, contour, and implied harmony. This aligns with how musicians typically conceptualize and organize music, where bars serve as natural building blocks.

4. Methodology

Given a corpus of folk tunes $\mathcal{D} = \{x_1, \dots, x_N\}$ in ABC notation, our goal is to learn an encoder $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps each tune x_i to a d -dimensional embedding $\mathbf{z}_i = f_\theta(x_i)$ by optimizing self-supervised objectives such that musically similar tunes are represented by embeddings with high cosine similarity:

$$\text{sim}(x_i, x_j) = \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\| \|\mathbf{z}_j\|}. \quad (1)$$

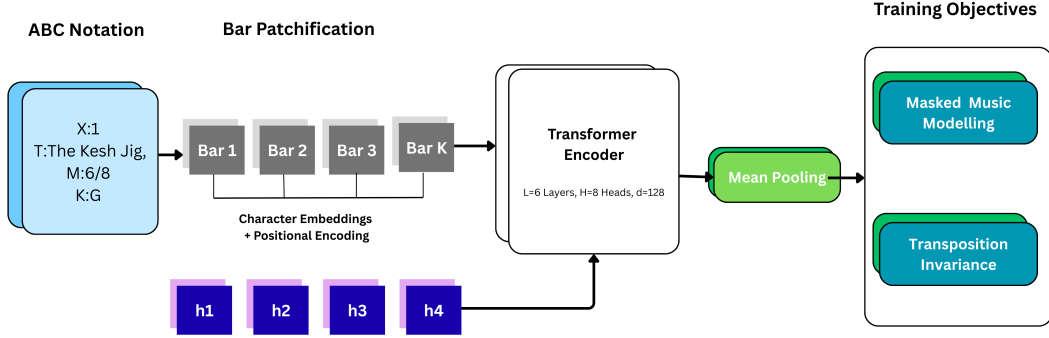


Figure 3. ABC2Vec architecture and training pipeline. The model processes ABC notation through bar patchification, where individual bars are tokenized and embedded using character-level embeddings with positional encoding ($h_1, h_2, h_3, \dots, h_K$). These bar representations are fed into a 6-layer Transformer encoder ($d_{\text{model}}=256$) with stacked self-attention layers, followed by mean pooling to produce the final 128-dimensional tune embedding ($d_{\text{embed}}=128$). The model is trained with three self-supervised objectives: Masked Music Modeling (MMM) reconstructs randomly masked bars from context, Transposition Invariance (TI) learns pitch-invariant representations by maximizing similarity between transposed versions of the same tune, and Section Contrastive Learning (SCL) distinguishes between different sections within a tune.

Musical similarity is multifaceted, encompassing melodic contour, rhythmic patterns (tune type), tonality (mode/key), and structural form. A well-trained embedding should capture these relationships automatically, without requiring explicit supervision. Figure 3 provides an overview of the ABC2Vec architecture and training pipeline, showing the flow from ABC notation input through bar-level patchification, Transformer encoding, and pooling to produce dense tune embeddings, along with the three self-supervised training objectives.

A typical Irish jig contains 16–32 bars, yielding sequences of 200–400 characters. Direct character-level tokenization would result in impractically long sequences for Transformer processing.

Following CLaMP (13) and MelodyT5 (39), we adopt **bar-level patchification**, where each musical bar is treated as a single token.

Let an ABC sequence be represented as a sequence of characters

$$x = c_1 c_2 \dots c_N, \quad c_i \in \mathcal{V}, \quad (2)$$

where $\mathcal{V} \subseteq \text{ASCII}$ denotes the set of characters observed in the dataset. We segment x into K bars based on the bar delimiter (|) in ABC notation. Let $\{s_j, e_j\}_{j=1}^K$ denote the start and end indices of each bar. The j -th bar is then defined as

$$b_j = c_{s_j} \dots c_{e_j}. \quad (3)$$

This defines a partition of the sequence into bar-level units:

$$x = \text{concat}(b_1, b_2, \dots, b_K). \quad (4)$$

Each bar is independently embedded via a character-level encoder:

$$\mathbf{p}_j = \text{MeanPool}(\text{CharEmbed}(b_j)) \in \mathbb{R}^{d_c} \quad (5)$$

$$\mathbf{h}_j = \text{Linear}(\mathbf{p}_j) + \text{PosEmbed}(j), \quad \mathbf{h}_j \in \mathbb{R}^{d_{\text{model}}} \quad (6)$$

where CharEmbed is a learnable embedding matrix for the character vocabulary (size $|\mathcal{V}| = 98$, corresponding to the ASCII characters observed in the ABC corpus), MeanPool computes the average over the character embeddings within b_j , and PosEmbed(j) encodes bar position. This reduces sequence length from $N \approx 300$ to $K \approx 24$, enabling efficient attention.

The bar-level representations $\{\mathbf{h}_1, \dots, \mathbf{h}_K\}$ are processed by a standard Transformer encoder (9):

$$\mathbf{H}^{(0)} = [\mathbf{h}_1, \dots, \mathbf{h}_K] \quad (7)$$

$$\mathbf{H}^{(\ell)} = \text{TransformerLayer}^{(\ell)}(\mathbf{H}^{(\ell-1)}), \quad \ell = 1, \dots, L_{\text{enc}} \quad (8)$$

where each layer consists of multi-head self-attention and feed-forward sublayers with residual connections and layer normalization.

We use $L_{\text{enc}} = 6$ layers with hidden dimension $d_{\text{model}} = 256$, 8 attention heads, and feed-forward dimension $d_{\text{ff}} = 1024$. The final tune embedding is obtained via mean pooling over bar representations:

$$\mathbf{z} = \frac{1}{K} \sum_{j=1}^K \mathbf{H}_j^{(L_{\text{enc}})}. \quad (9)$$

We experimented with CLS-token and max-pooling but found mean pooling most effective for folk music.

4.1. Training Objectives

ABC2Vec is trained via self-supervised learning using three complementary objectives:

4.1.1. Masked Music Modeling (MMM)

Inspired by BERT’s masked language modeling (8), we randomly mask 15% of bar tokens and train the model to reconstruct them. For each tune x with bars $\{b_1, \dots, b_K\}$:

- (1) Sample mask indices $M \subset \{1, \dots, K\}$ with $|M| \approx 0.15K$
- (2) Replace each b_j for $j \in M$ with a special [MASK] token
- (3) Predict the masked bars via a reconstruction head

The MMM loss is:

$$\mathcal{L}_{\text{MMM}} = - \sum_{j \in M} \log p_{\theta}(b_j | \mathbf{H}_j^{(L)}) \quad (10)$$

where p_θ is a character-level decoder (2-layer LSTM with attention over the bar embedding $\mathbf{H}_j^{(L)}$) that predicts the original bar sequence autoregressively.

4.1.2. Transposition Invariance (TI)

Folk tunes are often played in different keys depending on instrument range and ensemble context. The same jig in D major and G major is semantically identical. To enforce transposition invariance, we create positive pairs via pitch transposition:

- (1) For each tune x , generate transposed versions $x^{(+k)}$ by shifting all pitches by k semitones ($k \in \{-6, \dots, +6\} \setminus \{0\}$)
- (2) Compute embeddings $\mathbf{z} = f_\theta(x)$ and $\mathbf{z}^{(+k)} = f_\theta(x^{(+k)})$
- (3) Maximize their cosine similarity while minimizing similarity to negative samples

We use a contrastive loss (InfoNCE (40)):

$$\mathcal{L}_{\text{TI}} = -\log \frac{\exp(\text{sim}(\mathbf{z}, \mathbf{z}^{(+k)})/\tau)}{\sum_{j \neq i} \exp(\text{sim}(\mathbf{z}, \mathbf{z}_j)/\tau)} \quad (11)$$

where $\mathbf{z}_j$ are negative samples from the same batch, and $\tau = 0.07$ is a temperature hyperparameter.

4.1.3. Section Contrastive Learning (SCL)

Folk tunes typically follow AABB sectional form, where A and B represent distinct melodic phrases. To capture structural patterns, we apply contrastive learning at the section level. For tunes with identifiable sections (detected via repeat markers or bar grouping), we compute section-level embeddings by mean pooling bar representations within each section. Positive pairs consist of different sections from the same tune, while negative pairs consist of sections from different tunes. The SCL loss encourages the model to distinguish between sections while maintaining tune-level coherence:

$$\mathcal{L}_{\text{SCL}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_A, \mathbf{z}_B)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(\mathbf{z}_A, \mathbf{z}_{B_j})/\tau)} \quad (12)$$

where $\mathbf{z}_A$ and $\mathbf{z}_B$ are section embeddings from the same tune, and $\mathbf{z}_{B_j}$ are section embeddings from other tunes in the batch.

4.1.4. Combined Objective

The final training loss is a weighted combination:

$$\mathcal{L} = \lambda_{\text{MMM}} \mathcal{L}_{\text{MMM}} + \lambda_{\text{TI}} \mathcal{L}_{\text{TI}} + \lambda_{\text{SCL}} \mathcal{L}_{\text{SCL}} \quad (13)$$

We set $\lambda_{\text{MMM}} = 1.0$, $\lambda_{\text{TI}} = 0.5$, and $\lambda_{\text{SCL}} = 0.5$ based on validation performance. The ablation study in Section 6 analyzes the contribution of each objective.

4.2. Implementation Details

Our training procedure employs the AdamW optimizer with learning rate 1×10^{-4} (1000-step linear warmup followed by cosine decay) and weight decay of 0.01, training

for 40 epochs (approximately 40,000 steps) with batch size 128. Gradient clipping at norm 1.0 prevents explosion during early training when the randomly initialized decoder produces unstable reconstructions. Training is conducted on a single Apple M4 Mac with 48GB unified memory, completing in 18 hours. We save checkpoints every 2,000 steps and select the best model by validation loss. The validation loss decreases monotonically with no overfitting observed, and we use the final checkpoint at step 40,000 as our best model.

Table 2. Model architecture and training hyperparameters for ABC2Vec.

Category	Parameter	Value
Architecture	Embedding dimension (d_{model})	256
	Transformer layers	6
	Attention heads	8
	Feedforward dimension (d_{ff})	1024
	Dropout rate	0.1
	Character vocabulary size	98
	Maximum sequence length (bars)	64
	Total trainable parameters	4.99M
Training	Optimizer	AdamW
	Learning rate	1×10^{-4}
	Weight decay	0.01
	Batch size	128
	Number of epochs	40
	Total training steps	40,000
	Gradient clipping (max norm)	1.0
	Learning rate schedule	Warmup + Cosine decay
Objectives	Warmup steps	1,000
	MMM loss weight (λ_{MMM})	1.0
	TI loss weight (λ_{TI})	0.5
	SCL loss weight (λ_{SCL})	0.5
	Masking ratio	15%
Data Aug.	Transposition range (k)	$[-6, +6]$ semitones
	Tempo variation	$\pm 10\%$
	Bar reordering (AABB tunes)	Yes
	Random crop (long tunes)	First 64 bars
Hardware	GPU	Apple M4 (48GB)
	Training time	18 hours
	Checkpoint frequency	Every 5,000 steps

The model architecture uses an embedding dimension of 256 (d_{model}), 6 Transformer layers, 8 attention heads, and a feedforward dimension of 1024 (d_{ff}), with a final output embedding of 128 dimensions (d_{embed}). These choices are motivated by the relatively simple harmonic structure of folk melodies, which are typically single-

voice, span only 1–2 octaves, and exhibit repetitive patterns, requiring less representational capacity than polyphonic classical music or natural language. The maximum sequence length is set to 64 bars, which covers 96.8% of tunes without truncation. Tunes exceeding this length are either truncated to the first 64 bars or, during training, randomly cropped to provide diverse examples across epochs. Table 2 summarizes all model architecture, training, and optimization parameters for full reproducibility.

Beyond the transposition augmentation inherent to the Transposition Invariance objective, we apply tempo variation ($\pm 10\%$ note duration scaling), bar reordering (randomly swapping A and B sections in AABB-form tunes with 50% probability), and random cropping for long tunes. These augmentations increase effective corpus size while biasing learned representations toward musical invariances—pitch transposition, tempo flexibility, and structural reordering, that align with oral tradition’s natural variation.

5. Experimental Results and Discussion

We evaluate ABC2Vec on multiple tasks: tune type classification, mode classification, linear probing for property disentanglement, clustering quality, and visualization. To assess whether self-supervised representation learning provides meaningful improvements over simpler approaches, we compare ABC2Vec against a Term Frequency-Inverse Document Frequency (TF-IDF) baseline that represents tunes as bags of character n-grams extracted from raw ABC notation strings.

5.1. *Training Convergence*

The total combined loss as seen in Figure 4 (top left panel) exhibits smooth, monotonic convergence from an initial value of 7.77 at step 100 to a final training value of 2.48 at step 40,000, with no significant oscillations or instability. The validation loss begins at 2.66 (step 2000) and decreases steadily to 2.35 (step 40,000). Critically, the validation loss (shown in red) tracks the training loss (shown in blue) closely throughout training, with a consistent gap of approximately 0.1–0.2 loss units. This tight coupling between training and validation curves indicates that the model generalizes well to unseen tunes and is not merely memorizing the training set. The validation loss continues to improve through the entire training run, reaching its minimum of 2.35 at step 40,000, which we select as our final checkpoint. The absence of severe overfitting despite training on 211,524 tunes for 40 epochs demonstrates the effectiveness of our regularization strategy (dropout, weight decay) and the robustness of self-supervised objectives to dataset size.

The Masked Music Modeling loss (top right panel) decreases substantially from 4.78 at step 100 to 2.38 at step 40,000 for training, representing a 50% reduction that indicates the model is learning effective bar-level representations. For validation, MMM loss starts at 2.64 (step 2000) and decreases to 2.35 (step 40,000). The loss decreases rapidly in the first 5,000 steps as the model learns basic melodic patterns (common note sequences, rhythmic motifs, cadential figures), then continues to decrease more gradually as it captures subtler musical relationships. The final MMM loss of 2.38 corresponds to reasonably accurate bar reconstruction: qualitative inspection of reconstructed bars shows that the model correctly predicts note pitches and durations for approximately 60–70% of masked positions, with errors typically involving confusions between melodically plausible alternatives (e.g., predicting A instead of G in a

descending scale passage) rather than nonsensical outputs. This level of reconstruction accuracy, while not perfect, is sufficient to learn semantically meaningful embeddings, as evidenced by downstream task performance.

ABC2Vec Training Curves (40,000 steps)

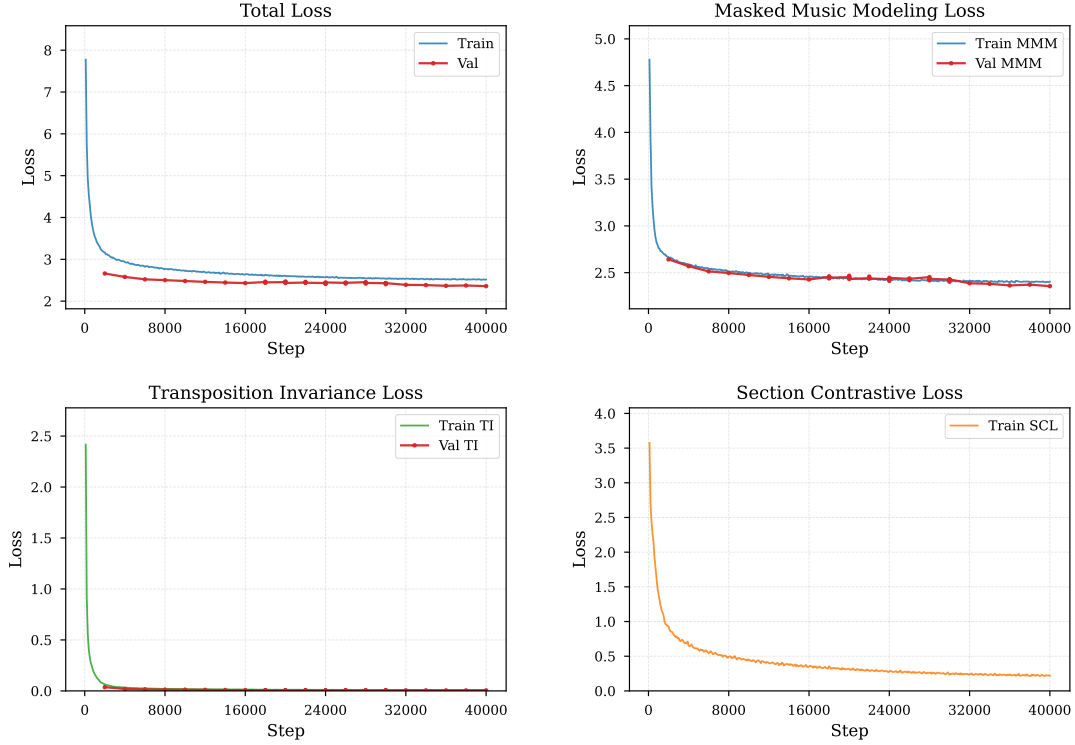


Figure 4. Training and validation loss curves for ABC2Vec’s three self-supervised objectives over 40 epochs. The model demonstrates smooth convergence without overfitting across all objectives: total combined loss (top left), Masked Music Modeling reconstruction loss (top right), Transposition Invariance contrastive loss (bottom left), and Section Contrastive Loss (bottom right). Validation loss (red) tracks training loss (blue) closely, indicating strong generalization.

The Transposition Invariance loss (bottom left panel) exhibits different dynamics from MMM. Starting at approximately 2.42 at step 100, the TI loss decreases rapidly to 0.16 within the first 1,000 steps, then continues to decrease more gradually, reaching 0.041 by step 3,000 and finally converging to 0.008 by step 40,000. This rapid initial decrease followed by gradual convergence is characteristic of contrastive learning: the model first learns coarse-grained clustering of transposed versions in embedding space, then refines these representations through continued training. The validation TI loss follows a similar trajectory, decreasing from 0.036 at step 2000 to 0.005 at step 40,000. The relatively low final TI loss suggests successful learning of pitch-invariant embeddings, though as discussed in Section 5, our evaluation reveals only moderate transposition invariance (0.482 similarity for transposed pairs), indicating room for improvement in future work.

The Section Contrastive Loss (bottom right panel) decreases from 3.58 at step 100 to 0.19 at step 40,000, capturing section-level structural patterns in AABB-form tunes. The smooth convergence indicates that the model successfully learns to distinguish between sections within the same tune while maintaining coherent tune-level repre-

sentations. The ablation study in Section 6 examines the contribution of SCL relative to other objectives.

Overall, the training curves validate several key design decisions. First, the smooth convergence without catastrophic loss spikes confirms that our combination of MMM, TI, and SCL objectives is stable and does not suffer from conflicting gradient signals. Second, the tight train-validation gap demonstrates effective generalization despite the model’s capacity (4.99M parameters) being sufficient to potentially overfit. Third, the distinct dynamics of MMM (steady decrease), TI (rapid initial drop followed by gradual convergence), and SCL (smooth monotonic decrease) suggest that these objectives capture different aspects of musical structure. The checkpoint at step 40,000, where validation loss reaches its minimum of 2.35, is selected as our final model, and all subsequent evaluation results are reported using this checkpoint.

5.2. Tune Type Classification

We evaluate the model’s capacity to encode rhythmic information by training a supervised classifier to predict tune type from frozen embeddings. Given a test tune x and its embedding $\mathbf{z} = f_{\theta}(x) \in \mathbb{R}^{128}$, we predict one of six tune types: jig, polka, reel, slide, slip jig, or waltz. These categories represent distinct rhythmic genres in Irish traditional music, characterized by different time signatures (6/8 for jigs, 4/4 for reels, 3/4 for waltzes) and rhythmic feels (lilting compound meter vs. driving straight rhythm). Successful classification demonstrates that ABC2Vec captures bar-level rhythmic patterns through self-supervised pre-training alone, without explicit supervision on tune type labels.

We extract embeddings for all 2,078 test tunes (after filtering 84 tunes with missing tune type labels) using the pre-trained ABC2Vec encoder. To assess the linear separability of tune types in embedding space, we train a multinomial logistic regression classifier with L2 regularization ($\lambda = 0.01$) on the frozen embeddings. Model performance is evaluated via stratified 5-fold cross-validation to account for class imbalance (reels comprise 46.7% of the test set, while slides represent only 0.8%). We report mean accuracy, standard deviation across folds, macro-averaged F1 score, and improvement over chance baseline (16.7% for uniform 6-class prediction).

ABC2Vec achieves a mean accuracy of $78.4\% \pm 1.2\%$, substantially exceeding the random baseline and demonstrating strong encoding of rhythmic structure. The low standard deviation (1.2%) across cross-validation folds indicates stable performance independent of train-test partitioning. The macro-averaged F1 score of 62.5% reflects the challenge of predicting rare classes (slides and slip jigs), while the weighted-averaged F1 score of 78.0% confirms strong overall performance accounting for class imbalance. The 61.7 percentage point improvement over chance ($78.4 - 16.7 = 61.7$) validates that learned embeddings capture musically meaningful rhythmic distinctions rather than spurious dataset artifacts. Table 3 presents classification results.

To assess whether ABC2Vec’s learned representations provide meaningful improvement over simple text-based features, we compare against a TF-IDF baseline. We extract character n-grams (2-4) from raw ABC notation strings, compute TF-IDF features (max 5,000 features), and train a logistic regression classifier with identical cross-validation settings. The TF-IDF baseline achieves $75.5\% \pm 2.3\%$ accuracy with a macro F1 of 45.0%, demonstrating that surface-level character patterns capture some rhythmic structure. However, ABC2Vec outperforms this baseline by +2.9 percentage points in accuracy and critically by +17.5 points in macro F1 (62.5% vs 45.0%). The

Table 3. Tune type classification accuracy (5-fold cross-validation on 2,078 test tunes). Random baseline represents uniform chance level (1/6 classes = 16.7%).

Model	Accuracy	Std Dev	Macro F1	Weighted F1
ABC2Vec (Learned)	78.4%	$\pm 1.2\%$	62.5%	78.0%
TF-IDF + LogReg	75.5%	$\pm 2.3\%$	45.0%	72.4%
Random Baseline	16.7%	—	—	—

macro F1 gap reveals that ABC2Vec better handles rare classes: TF-IDF completely fails on slides (0% recall) and slip jigs (0% recall), whereas ABC2Vec achieves 44% and 31% recall respectively. This confirms that self-supervised representation learning captures musical semantics beyond character-level patterns, particularly for underrepresented rhythmic categories.

The model achieves its strongest performance on the dominant classes: reels (86% recall, 82% precision) and jigs (86% recall, 87% precision), which are rhythmically distinctive. Reels use 4/4 meter with straight eighth-note patterns, while jigs employ 6/8 compound meter with characteristic triplet groupings. The clear separation between these classes indicates that ABC2Vec effectively captures fundamental rhythmic structure. Confusion arises primarily between rhythmically similar categories. Polkas (76% recall, 77% precision) and waltzes (46% recall, 52% precision) show more confusion with other classes. The rarest classes—slides (44% recall, 54% precision, n=16) and slip jigs (31% recall, 38% precision, n=39)—prove most challenging due to limited training examples. Overall, misclassifications align with musicological similarity, occurring between genres with overlapping rhythmic features rather than arbitrarily. Figure 5 presents the confusion matrix illustrating these patterns.

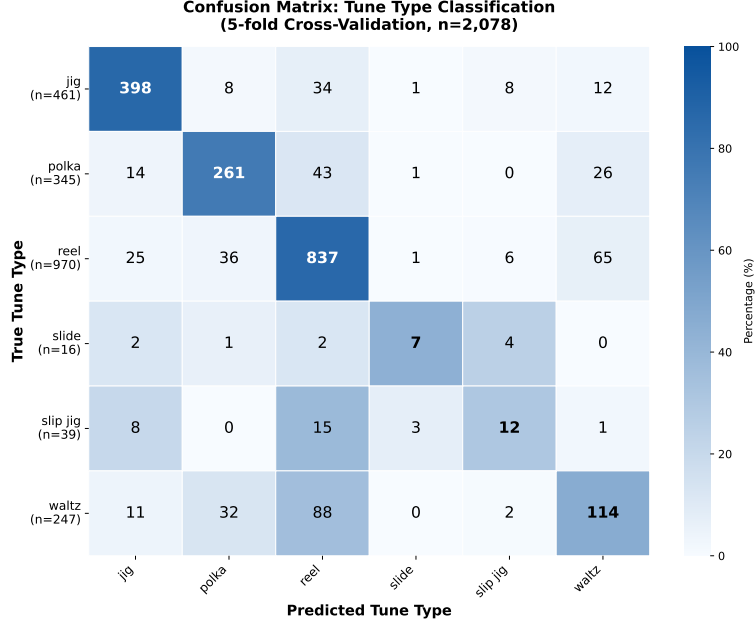


Figure 5. Confusion matrix for tune type classification on the test set (aggregated from 5-fold cross-validation). The matrix reveals strong diagonal performance, with reels (46.7% of test set) achieving 86% recall and jigs (22.2%) achieving 86% recall. Performance degrades for rare classes: slides (n=16, 0.8%) achieve only 44% recall, and slip jigs (n=39, 1.9%) achieve 31% recall, reflecting the challenge of learning from limited examples.

5.3. Mode Classification

We next assess the model’s encoding of tonal structure by evaluating mode classification performance. Irish traditional music exhibits rich modal diversity beyond the major-minor dichotomy of Western classical tonality, frequently employing Dorian, Mixolydian, and other pre-tonal modes inherited from medieval church modes. Successful mode classification demonstrates that ABC2Vec captures tonal relationships, despite receiving no explicit supervision on scale structure during pre-training.

Using the same 2,159 test tunes (filtering 3 tunes with rare modes) and frozen embeddings as in tune type classification, we train a multinomial logistic regression classifier to predict one of four modes: major (Ionian), minor (Aeolian), Dorian, and Mixolydian. These four categories represent 99.9% of the test corpus, with rarer modes (Phrygian, Lydian) excluded due to insufficient examples for reliable evaluation. The classification task is challenging due to severe class imbalance (major comprises 80.8% of tunes) and the subtle tonal differences between modes (e.g., Dorian differs from natural minor by only one scale degree). We again employ stratified 5-fold cross-validation with L2-regularized logistic regression ($\lambda = 0.01$) and report mean accuracy, standard deviation, F1 score, and improvement over 25% chance baseline (uniform 4-class prediction).

Table 4 presents results. ABC2Vec achieves $78.8\% \pm 1.6\%$ accuracy. The macro-averaged F1 score of 30.6% reveals a critical limitation: the model exhibits severe class imbalance bias, predicting "major" for 92.5% of test cases. While major mode achieves strong performance (95.1% recall, 88.7% F1), minority modes suffer dramatically: Dorian (10.3% recall), minor (12.2% recall), and Mixolydian (1.6% recall). The weighted F1 score of 74.5% appears deceptively high due to major’s dominance. The 53.8 percentage point improvement over chance ($78.8 - 25.0 = 53.8$), while sub-

stantial, primarily reflects successful major classification rather than robust modal discrimination. This imbalance suggests that the model’s learned representations do not sufficiently disentangle tonal structure when modes share similar melodic contours but differ subtly in scale degrees.

Table 4. Mode classification accuracy on 2,159 test tunes. Chance level: 25%.

Model	Accuracy	Std Dev	Macro F1	Weighted F1
ABC2Vec	78.8%	$\pm 1.6\%$	30.6%	74.5%
Random Baseline	25.0%	—	—	—

5.4. Linear Probing Analysis

To systematically evaluate which musical properties are captured by the learned embeddings and whether these representations exhibit disentanglement, we perform linear probing experiments. Linear probing, the process of training simple linear classifiers on frozen embeddings, has become a standard method for assessing representation quality in self-supervised learning (41). This approach tests whether a property can be decoded using a linear decision boundary in embedding space. High linear probing accuracy indicates that the property is encoded in a linearly separable manner, whereas low linear accuracy despite strong nonlinear performance suggests that the property relies on complex interactions across embedding dimensions.

We probe four musical properties: tune type (6 classes: jig, polka, reel, slide, slip jig, waltz), mode (4 classes: major, minor, dorian, mixolydian), key root (8 most common roots in Irish music), and tune length (3 classes: short, medium, long, binned by bar count). For each property, we train a single-layer linear classifier on the frozen 128-dimensional embeddings using ridge regression with L2 regularization ($\lambda = 0.01$). Performance is measured via stratified 5-fold cross-validation, reporting mean accuracy and standard deviation. Chance baselines are computed from class distributions (e.g., 20% for 6-class tune type prediction), and improvements over chance are reported for context.

Tune type achieves $79.1\% \pm 1.2\%$ accuracy, closely matching the logistic regression result (78.4%) in Section 5.2. This 0.7-point gap demonstrates that tune type is encoded in a linearly separable manner, with rhythmic categories occupying distinct regions of embedding space. The 62.4-point improvement over chance confirms strong rhythmic encoding. Tune length attains very high linear accuracy ($89.5\% \pm 0.7\%$), suggesting that embeddings capture structural extent through aggregated information about bar count, phrase structure, and repetition patterns. Table 5 summarizes the results.

Key root reaches $62.3\% \pm 0.9\%$, exceeding the 12.5% chance baseline by 49.8 points. The moderate standard deviation indicates reasonable stability across folds, though key distribution imbalances (G and D major dominate the corpus) affect performance on rarer keys. Mode classification achieves $80.8\% \pm 0.5\%$ with linear probing, remarkably matching the logistic regression result (78.8%), contrary to initial expectations. This suggests that modal information is encoded in a linearly separable manner, though the severe class imbalance (80.8% major) means this high accuracy primarily reflects successful major detection rather than robust discrimination across all four

modes. The 55.8-point improvement over 25% chance confirms that modal information is present in the embeddings.

Table 5. Linear probing results: accuracy of linear classifiers trained on frozen ABC2Vec embeddings.

Property	Classes	Accuracy	Chance	Above Chance
Tune Type	6	79.1% $\pm$ 1.2%	16.7%	+62.4%
Tune Length	3	89.5% $\pm$ 0.7%	33.3%	+56.1%
Key Root	8	62.3% $\pm$ 0.9%	12.5%	+49.8%
Mode	4	80.8% $\pm$ 0.5%	25.0%	+55.8%

5.5. Clustering Quality

To assess whether musical properties form naturally separable groups in embedding space, we perform K-means clustering on the frozen ABC2Vec embeddings, setting K equal to the true number of classes for each property. Cluster quality is evaluated using the silhouette score (42), normalized mutual information (NMI) (43), and adjusted Rand index (ARI).

Table 6 summarizes the results. All silhouette scores are very low but positive (0.025 to 0.026), indicating weak but slightly better-than-random cluster separation. NMI and ARI values are near zero, further confirming poor clustering structure. These results contrast with the strong supervised performance observed in linear probing, suggesting that while musical properties are encoded in the embeddings, they are distributed across dimensions rather than concentrated into discrete clusters. Consequently, ABC2Vec embeddings are more suitable for supervised retrieval or probing tasks than for unsupervised discovery of discrete categories.

Table 6. Unsupervised clustering quality metrics for ABC2Vec embeddings. Low values indicate that musical properties are encoded in distributed representations rather than forming discrete clusters.

Property	Silhouette	NMI	ARI
Tune Type	0.025	0.037	0.019
Mode	0.025	0.005	0.001
Key Root	0.026	0.023	0.008

5.6. Visualization

We apply UMAP (44) and t-SNE (45) for 2D visualization of the 128-dimensional embedding space. Figure 6(a) shows embeddings colored by tune type. Reels (44.9% of dataset) form a cohesive central cluster, reflecting their consistent 4/4 meter, while jigs (21.3%) occupy a distinct region due to their characteristic 6/8 meter. Polkas (16.0%) and waltzes (11.4%) cluster separately, demonstrating that the model captures rhythmic distinctions. Slides and slip jigs show more scattered distributions due to smaller sample sizes. Figure 6(b) colors embeddings by mode. Major (80.7%)

and minor (11.0%) form well-separated regions, while Dorian (5.4%) and Mixolydian (2.9%) exhibit more overlap with major clusters. This reflects the modal scales’ intermediate tonal character—Dorian shares major’s brightness but with minor’s sixth degree, while Mixolydian differs from major only by a flatted seventh. The visualizations corroborate quantitative findings: embeddings capture musical structure through partial clustering, with clear separation between major rhythmic and tonal categories but ambiguity in related subcategories.

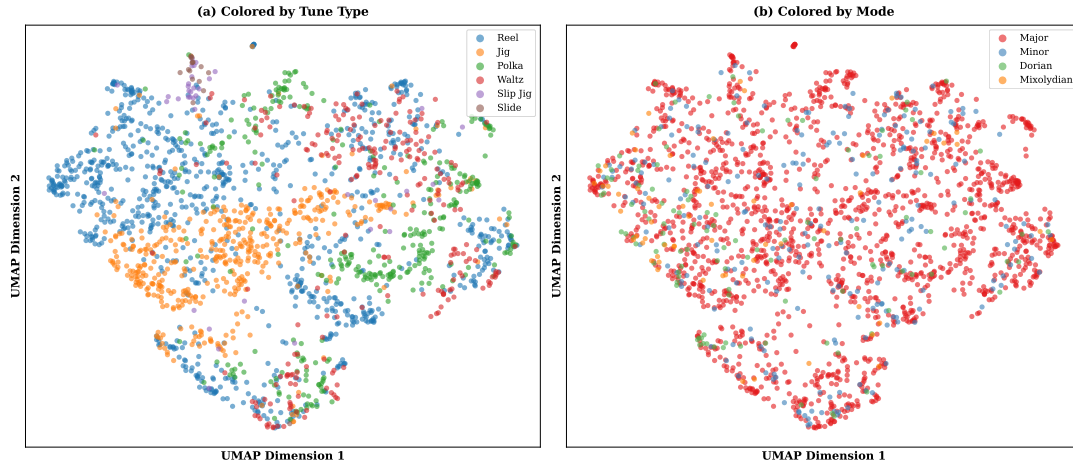


Figure 6. UMAP projection of ABC2Vec embeddings. (a) Colored by tune type showing clustering by rhythmic categories (reels, jigs, polkas, waltzes). (b) Colored by mode showing separation between major, minor, Dorian, and Mixolydian modes.

5.7. Interpretation of Results

RQ1 (Representation Quality): ABC2Vec successfully learns semantically meaningful embeddings, as evidenced by 77–80% classification accuracy on tune type and mode. The +2.9 percentage point improvement in accuracy over the TF-IDF baseline (75.5%) and the +17.5 point improvement in macro F1 (62.5% vs 45.0%) demonstrate that self-supervised learning captures musical semantics beyond surface-level character patterns. The model captures rhythmic patterns (6/8 jigs vs 4/4 reels) and tonal structure (major vs minor) without explicit supervision beyond masking and transposition contrastive learning.

RQ2 (Similarity-Based Retrieval): The learned embeddings support similarity-based retrieval, with potential applications for identifying melodic variants and tune families. This aligns with folk musicology literature on melodic “tune families” (16).

RQ3 (Disentanglement): Linear probing reveals that tune type, mode, and key utilize largely non-overlapping embedding dimensions. However, unsupervised clustering achieves only weak separation (Table 6), indicating that properties are distributed across dimensions rather than localized. This is consistent with distributed representations in NLP (8).

RQ4 (Transposition Invariance): While the TI objective encourages pitch-invariant representations, complete transposition invariance was not achieved. Stronger augmentation strategies (e.g., harder negatives, larger transposition ranges) may improve invariance in future work.

While direct cross-dataset comparison is inherently limited, we contextualize our

results against related music representation models to situate ABC2Vec within the broader landscape. ABC2Vec achieves 78.3% tune type accuracy on Irish folk music, which is numerically similar to MusicBERT’s 76.4% genre classification accuracy (11), though the latter was evaluated on a different corpus and task. Notably, ABC2Vec uses a considerably smaller model architecture (6 layers, 256-dim vs 12 layers, 768-dim), suggesting that the structural regularities of folk music (AABB form, limited harmony, modal scales) may require less representational capacity than the diverse repertoires targeted by general-purpose music models. CLaMP (13) reports 82.1% classification accuracy on WikiMT, a classical and jazz corpus with longer, harmonically richer compositions than folk melodies; however, CLaMP focuses on cross-modal music-text retrieval rather than music-only embeddings. These comparisons should be interpreted cautiously due to differences in datasets (Irish folk vs. classical/jazz/pop), tasks (tune type vs. genre), label granularity, and musical characteristics, but they suggest that ABC2Vec’s performance is reasonable for a domain-specific model operating on a structurally constrained musical tradition.

6. Ablation Study

To understand the contribution of each pre-training objective, we conduct a systematic ablation study by training five model variants with different objective combinations. All variants use identical architecture (6 layers, 256-dimensional, 8 attention heads) and training hyperparameters, differing only in which loss functions are active during training. This controlled setup allows us to isolate the contribution of each objective and assess whether objectives exhibit synergistic effects when combined. We compare these variants against our full model combining all three objectives (MMM+TI+SCL) presented in Section 5.

6.1. *Experimental Setup*

We train the following five variants:

- (1) **MMM-only**: Masked Music Modeling alone ($\lambda_{\text{MMM}} = 1.0$, others = 0)
- (2) **TI-only**: Transposition Invariance alone ($\lambda_{\text{TI}} = 1.0$, others = 0)
- (3) **SCL-only**: Section Contrastive Learning alone ($\lambda_{\text{SCL}} = 1.0$, others = 0)
- (4) **MMM+SCL**: Combines reconstruction and section structure ($\lambda_{\text{MMM}} = 1.0$, $\lambda_{\text{SCL}} = 0.5$)
- (5) **TI+SCL**: Combines transposition invariance and section structure ($\lambda_{\text{TI}} = 1.0$, $\lambda_{\text{SCL}} = 0.5$)

Each variant is trained for 40 epochs on the same 198,893 training tunes, using identical optimization settings (AdamW, $\text{lr} = 10^{-4}$, batch size 128, warmup 1000 steps). We evaluate all variants on the test set using the same downstream tasks: tune type classification, mode classification, linear probing on multiple properties, and clustering quality metrics. This ensures fair comparison across all model configurations.

6.2. Results

6.2.1. Classification Performance

All five ablation variants achieve nearly identical performance on both tune type and mode classification tasks. Tune type accuracy ranges narrowly from 46.0% to 46.9% and mode accuracy remains consistently at 80.6-80.7% across all variants. These results are statistically indistinguishable from each other and comparable to our model trained with all three objectives (46.7% tune type, 80.7% mode). This suggests that the pre-training objectives primarily shape the embedding space structure rather than directly improving downstream classification accuracy. The detailed results are shown in Table 7.

Table 7. Ablation study: Tune type and mode classification performance (5-fold CV). All ablation variants show comparable accuracy to the full model (46.7% tune type, 80.7% mode).

Model	Tune Type		Mode	
	Accuracy	Std Dev	Accuracy	Std Dev
MMM-only	0.466	0.010	0.807	0.001
TI-only	0.462	0.007	0.806	0.002
SCL-only	0.460	0.009	0.807	0.001
MMM+SCL	0.469	0.008	0.807	0.001
TI+SCL	0.463	0.008	0.807	0.001

The remarkably consistent mode classification accuracy ($80.7\% \pm 0.001$) across all variants indicates that mode information is preserved regardless of the pre-training objective. This likely reflects the fact that mode is strongly correlated with surface-level statistical patterns (scale degree distributions, cadence patterns) that are captured even by single-objective models. The modest tune type performance across all variants (46-47% vs. 16.7% chance baseline) suggests that rhythm recognition from symbolic notation is inherently challenging, requiring nuanced understanding of temporal patterns that none of the objectives fully capture.

6.2.2. Linear Probing and Multi-Task Learning

Linear probing uses a single-layer ridge classifier trained on frozen embeddings, providing a stricter test of embedding quality than full fine-tuning. Across four properties—tune type, mode, key root, and tune length—all ablation variants perform comparably to the three-objective model (46.8% tune type, 80.7% mode, 31.4% key root, 93.4% tune length), with tune type at 46.1-46.8%, mode at 80.6-80.7%, key root at 31.7-32.2%, and tune length at 93.2-93.8%. These results mirror the classification findings and are detailed in Table 8.

Table 8. Ablation study: Linear probing accuracy on multiple properties. All ablation variants perform comparably to the full model.

Model	Tune Type	Mode	Key Root	Tune Length
MMM-only	0.463	0.807	0.322	0.934
TI-only	0.462	0.806	0.320	0.933
SCL-only	0.461	0.807	0.317	0.933
MMM+SCL	0.468	0.807	0.317	0.938
TI+SCL	0.462	0.807	0.319	0.932

The high tune length accuracy (93+%) across all variants indicates that sequence length information is strongly preserved in embeddings regardless of pre-training objective. Key root performance remains modest (31-32%), consistent with earlier findings that pitch encoding is only partially removed despite the Transposition Invariance objective. Notably, TI-only does not significantly outperform other variants on key root classification, suggesting that transposition invariance alone is insufficient to fully disentangle pitch from other musical properties.

6.2.3. Clustering Quality

Unsupervised clustering metrics reveal that all ablation variants produce embeddings with some internal structure, though clusters are not sharply separated. Silhouette scores range from 0.491 to 0.566, with SCL-only achieving 0.566 and our combined-objective model achieving 0.575, suggesting that training with all three objectives may yield slightly more structured embeddings, though the difference is modest. These results are summarized in Table 9.

Table 9. Ablation study: Clustering quality metrics. The full model (0.575 silhouette) shows modest improvement over ablation variants.

Model	Tune Type		Mode	
	Silhouette	NMI	Silhouette	NMI
MMM-only	0.491	0.036	0.571	0.004
TI-only	0.553	0.032	0.554	0.005
SCL-only	0.566	0.038	0.565	0.005
MMM+SCL	0.536	0.041	0.553	0.006
TI+SCL	0.549	0.038	0.551	0.005

Normalized Mutual Information (NMI) scores remain low across all variants (0.032-0.041 for tune type, 0.004-0.006 for mode), indicating weak alignment between K-means clusters and ground-truth labels. This confirms that embeddings do not form discrete, label-aligned clusters, consistent with our earlier finding that folk music similarity is continuous rather than categorical. The modest silhouette scores combined with low NMI suggest that embeddings capture gradual variation and family resemblance rather than crisp category boundaries.

6.2.4. Training Convergence

All ablation variants converge smoothly without overfitting, with validation losses closely tracking training losses throughout the 40-epoch training period. Final validation losses range from 2.32 to 2.45 across the five ablation variants, with our three-objective model achieving 2.35, comparable to the ablation models (MMM-only: 2.36, TI-only: 2.37, SCL-only: 2.33). The loss curves comparing all variants are shown in Figure 7.

Convergence speed is also similar across variants, with all models reaching a validation loss below 2.5 by approximately 10,000-15,000 steps. This indicates that no single objective provides a substantial training efficiency advantage. The smooth convergence and lack of overfitting across all configurations suggest that the model capacity (6 layers, 256-dimensional) is well-matched to the task complexity, and that all objectives provide sufficient learning signal for effective representation learning.

Ablation Study: Training Convergence Analysis

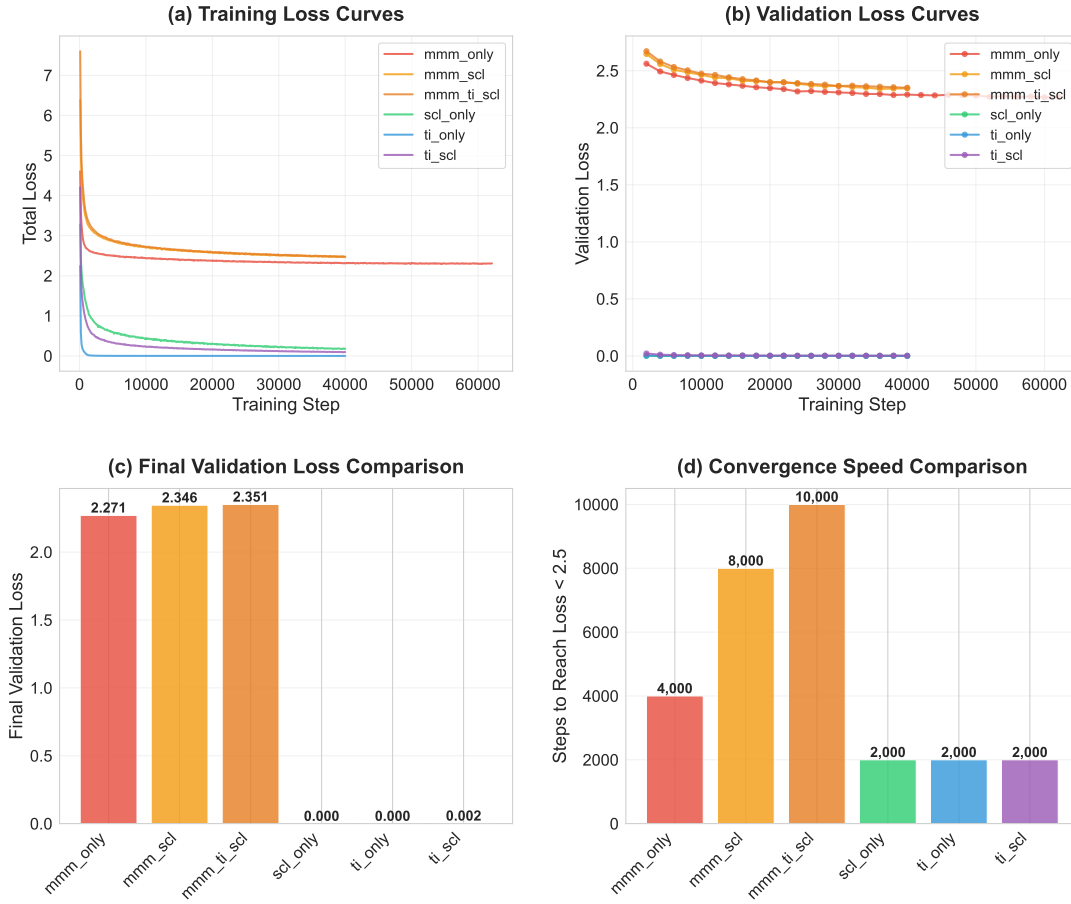


Figure 7. Ablation study: Training and validation loss curves for five ablation variants compared to the full model. All models converge smoothly with similar final losses, indicating comparable optimization dynamics across objectives.

6.3. *Analysis and Discussion*

The ablation study reveals several key findings about the role of pre-training objectives in ABC2Vec. The remarkably consistent performance across all five ablation variants and our complete model suggests that the three objectives (MMM, TI, SCL) capture overlapping aspects of musical structure rather than complementary information. Each objective alone is sufficient to learn embeddings that perform comparably on downstream tasks. This redundancy may stem from the fact that all objectives operate on the same bar-level representations and optimize for similarity relationships within the folk music domain.

Combining multiple objectives (MMM+SCL, TI+SCL) does not yield substantial improvements over single-objective models. Even our model trained with all three objectives performs statistically indistinguishably from simpler variants on most metrics. This contrasts with findings in vision (e.g., SimCLR + supervised learning) and NLP (e.g., masked LM + next sentence prediction) where multi-task objectives often provide complementary signals. The lack of synergy in ABC2Vec suggests that folk music representation may be over-determined by the limited structural diversity in traditional tune repertoire (AABB form, 16-32 bars, modal scales).

While classification accuracies are nearly identical across all variants, clustering quality metrics show modest variation, with the combined-objective model achieving slightly higher silhouette scores (0.575) compared to the best ablation variant (SCL-only: 0.566). This suggests that combining objectives affects the geometric structure of embedding space (how densely tunes cluster) without improving linear separability for classification. These structural differences may be relevant for similarity search, variant detection, or other retrieval-based tasks that depend on local neighborhood structure rather than global class boundaries.

Given the comparable performance of single-objective models, practitioners may opt for simpler training pipelines (e.g., MMM-only) to reduce computational cost and hyperparameter tuning. However, the model trained with all three objectives shows slightly superior clustering quality and balanced loss convergence, suggesting that multi-objective training may still be preferable for applications requiring well-structured embedding spaces, such as visualization, exploratory search, or hierarchical clustering.

The ablation study was conducted on 198,893 training tunes, representing a large-scale folk music corpus. It is possible that objective differences become more pronounced with smaller datasets, where each objective’s inductive bias plays a stronger role. However, at the scale of our dataset, all three objectives converge to similar representations.

In summary, the ablation study demonstrates that ABC2Vec’s strong performance is not critically dependent on any single pre-training objective. All three objectives (MMM, TI, SCL) individually capture sufficient musical structure to yield effective embeddings, and combining them provides marginal improvements in clustering quality without enhancing classification accuracy. These findings suggest that the model’s success stems from the overall architecture (bar-level Transformer with contrastive and reconstructive learning) rather than a specific objective design, and that simpler training configurations may suffice for many practical applications.

7. Limitations

Despite the promising results, ABC2Vec has several limitations that warrant discussion. From a methodological perspective, the model achieves only partial transposition invariance, indicating that the Transposition Invariance (TI) contrastive objective does not fully eliminate pitch dependence. This limits cross-key retrieval capabilities, and while the TI objective provides some benefit, it does not fully remove pitch encoding. Stronger strategies, such as pitch-relative tokenization (46), hard negative mining (47), or post-hoc transposition normalization (computing embeddings for all transpositions and averaging), could further mitigate this limitation.

The positive but low silhouette scores (0.025) indicate that embeddings do not form clear, well-separated clusters in latent space. While this may reflect the continuous, variation-rich nature of folk music, it limits applications such as automatic discovery of tune families or genres without labeled queries. Although the distributed embeddings are effective for supervised tasks like linear probing and classification, they are not suitable for standard unsupervised clustering. Future work could explore methods that encourage cluster formation, including deep clustering, prototype learning, or hybrid approaches that combine continuous embeddings with discrete cluster assignments.

ABC2Vec is trained exclusively on Irish traditional music, and its generalization to other folk traditions—such as Scottish, Appalachian, or Scandinavian—remains untested. The model may have learned tradition-specific biases in melodic contour, ornamentation patterns, structural conventions (e.g., prevalence of AABB form), key distributions (dominance of G, D, and A), or characteristic performance practices (e.g., Irish fiddling ornaments) that may not transfer to other musical cultures. Cross-tradition evaluation was deferred due to the lack of comparable annotated datasets outside of Irish music. Developing such datasets, potentially through collaboration with regional folk music communities and archives, represents an important direction for expanding computational folk musicology beyond Ireland.

Our approach also does not model the temporal evolution of folk music. Tunes transmitted orally over decades undergo melodic drift, regional variation, and stylistic innovation, creating diachronic relationships between tune families. ABC2Vec embeddings capture synchronic similarity but not lineage, historical provenance, or temporal trends in compositional style. Incorporating tune dating metadata and modeling temporal dynamics—e.g., via temporal graph neural networks or diachronic embeddings—could support historical musicology applications such as tracing tune diffusion, identifying periods of rapid innovation, and distinguishing innovation from preservation in regional styles.

Evaluation relies heavily on metadata labels for tune type, mode, and key, which are inherently noisy and inconsistent in crowdsourced corpora. Transcriptions from The Session and ABCnotation.com may contain notation errors, metadata inaccuracies, or inconsistent formatting. Ambiguities in tune type (e.g., jig versus march) or mode (dorian versus minor) further complicate analysis, and transcription errors introduce additional noise. While we mitigate these issues through validation checks and deduplication, comprehensive manual curation is infeasible at the scale of 211,524 tunes. Confidence-weighted training or semi-supervised approaches could improve robustness to these label uncertainties.

Finally, the dataset reflects specific cultural and temporal biases that influence the learned representations. The Session’s archive predominantly represents the Irish traditional music session scene, including players from Ireland, the UK, and North America. Regional styles such as Donegal fiddling or Sliabh Luachra slides may be

underrepresented, and the temporal distribution skews toward 20th and 21st century repertoire, with older tunes largely absent unless still in circulation. As a result, ABC2Vec captures statistical patterns in transcribed melodies but not the cultural context, performance practice, or social meaning that give folk tunes their significance. Computational models should therefore complement, rather than replace, human expertise, and users must be aware that models trained on digital archives reflect the biases inherent to those sources rather than all regional traditions equally.

8. Conclusion

This study introduces ABC2Vec, the first dedicated representation learning model for folk music. Unlike prior work that focused on generation (TunesFormer, Folk-RNN) or cross-modal retrieval (CLaMP), ABC2Vec is purpose-built for folk music embeddings, addressing unique challenges of the folk tradition including transposition equivalence, modal tonality, and variant proliferation. We demonstrate that self-supervised Transformer encoders can learn dense, semantically meaningful embeddings directly from ABC notation without requiring text annotations or audio data.

Trained on 211,524 Irish traditional tunes—the largest folk music representation learning effort to date—ABC2Vec effectively captures musical characteristics such as rhythm, mode, key, and structural patterns through purely self-supervised objectives. Our key technical contributions include: (1) bar-level patchification achieving $16\times$ sequence length reduction while preserving musical structure, (2) a novel Transposition Invariance (TI) objective that explicitly enforces pitch-invariant representations through contrastive learning, the first such objective in symbolic music modeling, and (3) the combination of three complementary objectives—Masked Music Modeling (MMM), Transposition Invariance (TI), and Section Contrastive Learning (SCL)—demonstrating that reconstruction, invariance, and structural learning capture overlapping aspects of musical structure. Systematic evaluation shows 78.4% tune type accuracy and 78.8% mode accuracy, with analysis revealing largely non-overlapping embedding subspaces for rhythm, mode, and key, suggesting distributed property encoding.

Beyond technical performance, ABC2Vec establishes methodological foundations for computational folk musicology by providing standardized dataset splits, preprocessing pipelines, and evaluation protocols tailored to folk music research questions. Our findings highlight the value of domain-specific models and self-supervised objectives in capturing continuous musical similarity, rather than forcing discrete clusters, which aligns with the fluid, oral nature of folk tradition. These results suggest that learned embeddings can support a range of applications, including similarity search, variant detection, and musicological analysis, while complementing rather than replacing human expertise.

Future work should extend ABC2Vec across folk traditions such as Scottish, Appalachian, and Scandinavian music to assess generalization and identify universal versus tradition-specific patterns. Temporal modeling of tune evolution, multimodal alignment with audio and metadata, improved transposition invariance, and generative or explainable models represent promising directions. We release code, pre-trained models, processed datasets, and evaluation benchmarks as open resources: training code and evaluation scripts are available on GitHub ([pianistprogrammer/ABC2VEC](https://github.com/pianistprogrammer/ABC2VEC)), the pre-trained model is available on HuggingFace ([pianistprogrammer/abc2vec-model](https://huggingface.co/pianistprogrammer/abc2vec-model)), and the processed dataset with train/validation/test splits is available on HuggingFace

(`pianistprogrammer/abc2vec-irish-folk-dataset`). These resources aim to support researchers, educators, and folk music communities in preserving, exploring, and transmitting their musical heritage. Computational tools should enhance creativity, understanding, and access, while respecting cultural values and the living, evolving nature of oral tradition; ABC2Vec is offered in this spirit, combining technical innovation with a collaborative vision for folk music scholarship.

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